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FILE SYSTEMS

All computer applications need to store and retrieve information. While a process is running, it can store a limited amount of information in physical RAM. For many applications, the amount of memory is far too small and some even need many terabytes of storage.

A second problem with keeping information in RAM is that when the process terminates, the information is lost. For many applications (e.g., for databases), the information must be retained for weeks, months, or even forever. Having it vanish when the process using it terminates is unacceptable. Furthermore, it must not go away when a computer crash kills the process or power goes off during an electrical storm.

A third problem is that it is frequently necessary for multiple processes to access (parts of) the information at the same time. If we have an online telephone directory stored inside the address space of a single process, only that process can access it, unless it is shared explicitly. The way to solve this problem is to make the information itself independent of any one process.

Thus, we have three essential requirements for long-term information storage:

1. It must be possible to store a very large amount of information.
2. The information must survive the termination of the process using it.
3. Multiple processes must be able to access the information at once.

Magnetic disks have been used for years for this long-term storage. While such disks are still used extensively, solid-state drives (SSDs) have also become hugely

popular, complementing or replacing their magnetic counterparts. Compared to hard disks, they do not have any moving parts that may break, and offer fast random access. Tapes and optical disks are no longer as popular as they used to be and have much lower performance. Nowadays, if they are used at all, it is typically for backups. We will study magnetic hard disks and SSDs more in Chap. 5. For the moment, you can think of both as *disk-like*, even though strictly speaking an SSD is not a disk at all. Here, “disk-like” means that it supports an interface that appears to be a linear sequence of fixed-size blocks and supporting two operations:

1. Read block k
2. Write block k

In reality there are more, but with these two operations one could, in principle, solve the long-term storage problem.

However, these are very inconvenient operations, especially on large systems used by many applications and possibly multiple users (e.g., on a server). Just a few of the questions that quickly arise are:

1. How do you find information?
2. How do you keep one user from reading another user’s data?
3. How do you know which blocks are free?

and there are many more.

Just as we saw how the operating system abstracted away the concept of the processor to create the abstraction of a process and how it abstracted away the concept of physical memory to offer processes (virtual) address spaces, we can solve this problem with a new abstraction: the file. Together, the abstractions of processes (and threads), address spaces, and files are the most important concepts relating to operating systems. If you really understand these three concepts from beginning to end, you are well on your way to becoming an operating systems expert.

Files are logical units of information created by processes. A disk will usually contain thousands or even millions of them, each one independent of the others. In fact, if you think of each file as a kind of address space, you are not that far off, except that they are used to model the disk instead of modeling the RAM.

Processes can read existing files and create new ones if need be. Information stored in files must be **persistent**, that is, not be affected by process creation and termination. A file should disappear only when its owner explicitly removes it. Although operations for reading and writing files are the most common ones, there exist many others, some of which we will examine below.

Files are managed by the operating system. How they are structured, named, accessed, used, protected, implemented, and managed are major topics in operating system design. As a whole, that part of the operating system dealing with files is known as the **file system** and is the subject of this chapter.